

# InterLock AI Review Console (Demo Project)

*AI-Assisted Cross-Document Review for Industrial Engineering Teams*

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## 1. Product

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InterLock AI Review Console is an AI-assisted document review tool for industrial energy infrastructure teams. It ingests engineering PDFs — owner specifications, control system specs, vendor submittals, standards references, and checklists — extracts structured technical claims, and surfaces verifiable discrepancies across documents with citations back to the exact source page and snippet.

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The system is designed as a traceable review assistant, not a compliance certifier. Every flag it generates is grounded in source evidence. Every extracted claim is inspectable. The system reduces manual review time and makes pre-procurement, pre-commissioning, and management-of-change reviews auditable.

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## 2. Reviewer User

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### PRIMARY USER

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Plant engineer, project engineer, HSE/safety lead, or design reviewer responsible for reviewing generator packages, control-system specifications, interlock logic, safety checklists, vendor submittals, and standards references before procurement or commissioning.

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### USER CONTEXT

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- Manages 5–30 PDF documents per project package: owner specs, vendor sheets, P&IDs, checklists, standards, and contracts.
- Discovers design errors late — often at commissioning — because no systematic cross-document check exists.
- Cannot trust an AI system that makes compliance claims without evidence. Needs to see exactly where a flag came from before acting on it.
- Works under regulatory scrutiny (OSHA, NEC, NFPA, IEC). Every review decision must be defensible and auditable.
- Time-constrained during design gates and procurement windows. Manual review of multi-document packages can take days.

### WHAT THE USER CURRENTLY DOES

Manually reads through PDFs, cross-referencing values by memory or annotation. Inconsistencies are found late or not at all. There is no systematic record of what was checked.

## 3. Problem

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Design errors in industrial energy infrastructure projects are routinely discovered late — at procurement, commissioning, or post-incident review — because critical technical assumptions are distributed across multiple PDF documents that are never systematically compared against each other.

A generator owner specification requires standby-rated equipment. The vendor submits a continuous-rated unit.

Nobody catches it because the owner spec is 47 pages and the vendor datasheet is a separate file.

The discrepancy surfaces at commissioning, after equipment is already on-site.

The specific failure modes this product addresses:

- Parameter mismatches: voltage, frequency, kW/kVA, power factor, runtime, alarm thresholds specified differently across documents.
- Assumption conflicts: standby vs. prime vs. continuous rating; indoor vs. outdoor enclosure; automatic vs. manual operation.
- Reference gaps: required standards (UL 2200, NFPA 110, IEC 61511) present in specs but absent from vendor submittals.
- Checklist gaps: required control features (remote emergency stop, battery low alarm, fault historian) specified in one document but not confirmed present in another.

Manual review is slow and brittle. It depends on reviewer memory and attention across dense technical documents. It produces no audit trail. It does not scale to the document volumes typical in plant engineering projects.

4. Workflow Fit

InterLock AI Review Console fits at decision gates where a document package is being reviewed before a consequential action:

| Review Stage         | What the System Surfaces                                                                   |
|----------------------|--------------------------------------------------------------------------------------------|
| Design gate review   | Parameter and assumption conflicts between owner spec and preliminary engineering drawings |
| Procurement approval | Discrepancies between project spec and vendor submittals before purchase order             |
| Commissioning review | Checklist gaps and standard reference mismatches before final acceptance testing           |

Management-of-change review

Parameter changes in revised documents that conflict with existing approved specs

**The product does not approve designs. It does not certify compliance. It surfaces evidence-backed review flags and recommends that an engineer verify before proceeding. This framing is deliberate: in safety-critical industrial systems, the human engineer retains judgment authority. The system's role is to reduce the probability of missed inconsistencies and to make the review record auditable.**

## 5. MVP

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### CORE WORKFLOW

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- User uploads 2–6 engineering PDFs and assigns document type (owner spec, control spec, vendor submittal).
- System parses each PDF page-by-page: text extraction via pdfplumber, table extraction, OCR fallback for scanned pages.
- System extracts structured claims: parameters (kW, V, Hz, PF), assumptions (rating type, enclosure, operation mode), standards references (UL, NFPA, IEC, IEEE), and checklist items.
- System detects discrepancies deterministically: numeric diff for parameters, enum conflict for assumptions, set arithmetic for references, presence check for checklist items.
- System adds cautious LLM-generated explanations to each flag. LLM only explains flags already found by the comparison engine — it does not detect them.
- User reviews discrepancy cards with full source citations: document name, page number, verbatim snippet, extraction method, confidence score.
- User exports an audit report (PDF or JSON) for project records.

### OUT OF SCOPE FOR MVP

- Vector database or semantic search over document corpus
- Role-based authentication or multi-user collaboration
- Agent orchestration or multi-step agentic workflows
- Compliance certification or regulatory filing
- "Ask a question" chatbot interface

## 6. Wedge-to-Platform Path

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### WEDGE

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**Generator and control package reviews for industrial facilities. Initial buyers: project engineers and HSE leads at industrial EPC firms, utilities, and plant operators reviewing generator packages before procurement.**

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### PLATFORM EXPANSION

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- Plant-wide interlock knowledge graph: build a live, queryable graph of all safety-instrumented functions, interlocks, and logic relationships across a facility's document library.
- Standard-specific compliance review: structured checkers for IEC 61511 SIL lifecycle, NFPA 110 generator system requirements, NEC electrical installations.
- Operator decision support: surface relevant engineering constraints at the control room during abnormal operating conditions.
- Management-of-change review: automatically diff revised documents against approved baselines and flag changed parameters.
- Regulator-grade audit trail: immutable, timestamped review record with citation provenance for regulatory inspection and incident investigation.
- OEM/submittal review marketplace: allow equipment vendors to pre-validate submittals against customer specs before submission.

*This demo project does not replace engineering judgment. It reduces the probability of missed inconsistencies and makes reviews auditable.*